AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 19 · STUDENT LESSON

Sampling and Representation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

AI-1 Week 22: repeated examples do not add representative variety.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Define a question and target population. Try the recall before opening notes.

Your learning target

I can define a question and target **population**.

I can describe how a **sample** was selected.

I can limit conclusions to supported groups.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 19: Sampling and **Representation**

A **sample** must reflect the people and situations where the system will be used.



THINKER MISSION

Audit a playground survey collected from one class.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

If 6 of 10 **response**s choose Reading, what is the observed proportion?

Readiness for Graphs and honest scales

Counts are 8 and 10. What is their absolute difference?

My readiness explanation

This week's end goal

By the end, I can define a **population**, selection frame and non**response** report for a survey.

Notice the evidence

> Investigate the supplied Sampling and Representation case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use sampling and **representation** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A fictional school wants to know which activity Grade 7 students prefer. Surveying only the coding club is easy, but the question concerns every Grade 7 student.

First define the **population**, selection **process**, question and **response** options. No real student data is required for this exercise.

Apply the connected idea: Graphs and honest scales

A fictional poll records 8 votes for Drawing and 10 for Building. A **bar chart** should make lengths proportional to counts, starting at zero.

A chart starting at 7 makes the visible bars 1 and 3 units tall and can exaggerate the difference.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Sampling and Representation. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

IN

INPUT

Audit a playground survey collected from one class.

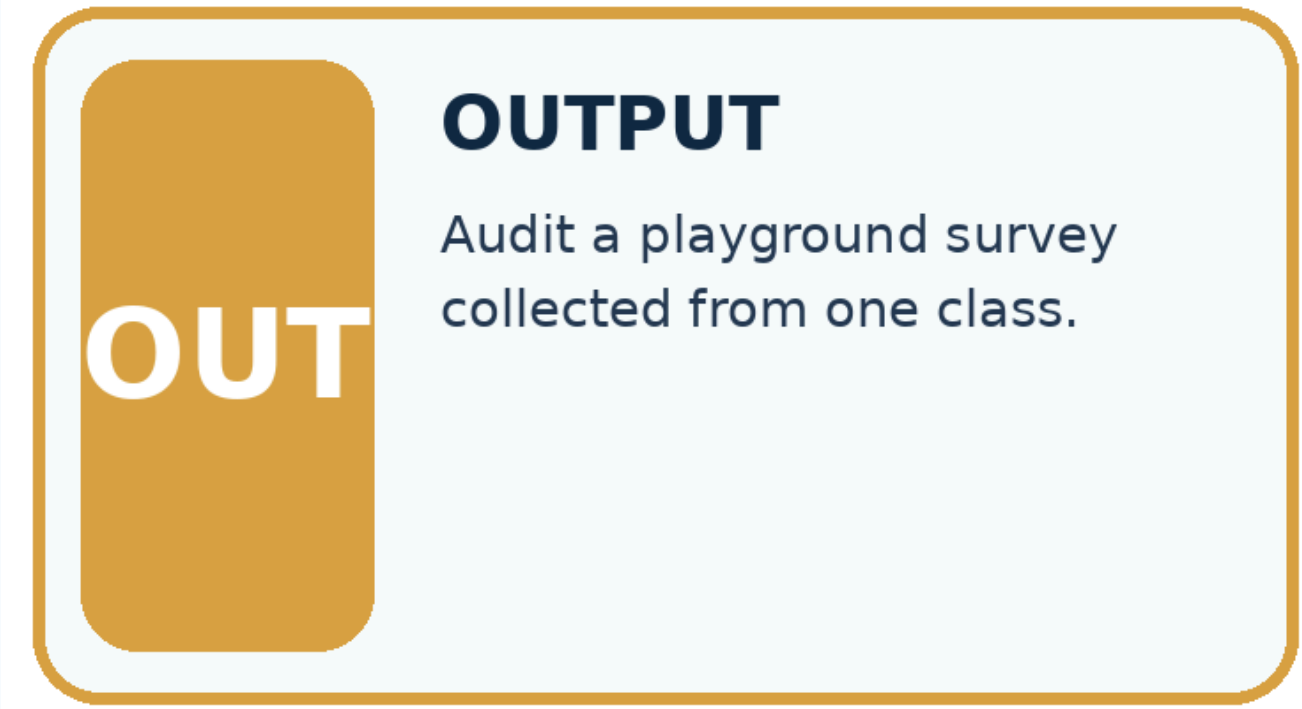
↓

↻

PROCESS

A sample must reflect the people and situations where the system will be used.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
rows = [{"group": "club", "choice": "Code"},
        {"group": "club", "choice": "Read"},
        {"group": "club", "choice": "Code"}]
code_votes = 0
for row in rows:
    if row["choice"] == "Code":
```

```
        code_votes = code_votes + 1
print(code_votes, len(rows))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Target	All Grade 7 students	Population
Available data	Three club responses	Sample
Observed result	2 of 3 choose Code	Describes respondents

Read each row from left to right and connect the evidence to the stated rule.

The result 2 of 3 describes these fictional club **response**s. It cannot by itself estimate the whole grade reliably. A better plan starts from a complete grade roster, uses a fair selection method and records non**response**.

Random selection gives each listed student a chance; it does not guarantee every small **sample** is representative.

Match the people to the question

The **population** is everyone the question concerns. The **sample** is the smaller group whose **response**s you actually collect.

If the question is about all Grade 7 students, a coding-club **sample** leaves other students out before the poll even starts. Separate the invitation **process** from the **response process**: some invited students may not reply.

A report needs both counts so readers can see whose preferences might be missing.

Apply the connected idea: Graphs and honest scales

```
counts = {"Drawing": 8, "Building": 10}
for label in counts:
    print(label, "#" * counts[label], counts[label])
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Drawing	########	8 votes
Building	##########	10 votes

Read each row from left to right and connect the evidence to the stated rule.

The text chart uses one # per vote and a zero **baseline**. The second category has 2 more votes and 1.25 times as many votes, not three times as many.

Include the poll question, date or collection occasion and number of **response**s when presenting a real graph.

Make bar lengths mean something

With a zero **baseline**, eight votes make a bar eight units high and ten votes make a bar ten units high. The difference is two votes.

If the drawing begins at seven, only the extra one and three units are visible, making the second bar look three times as tall. That picture describes the part above seven, not the full counts.

Use one **scale**, label the units and show the counts.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

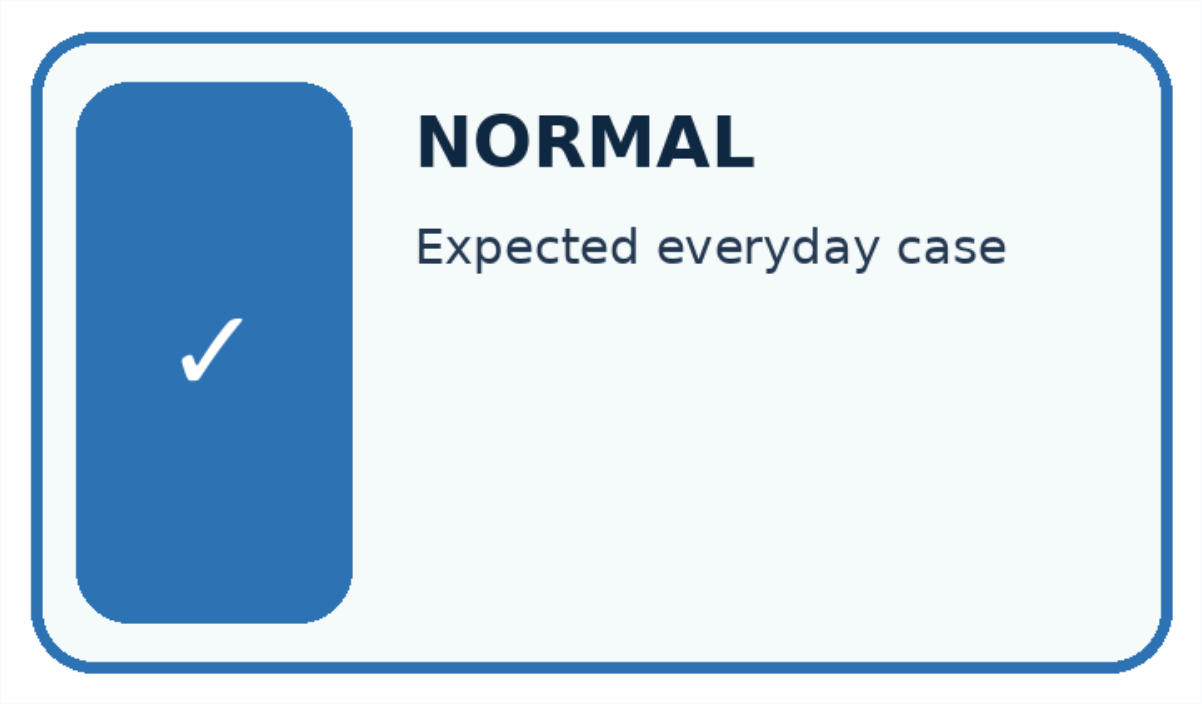
> Compare the supplied cases for Sampling and Representation. Identify the detail that changes the conclusion.

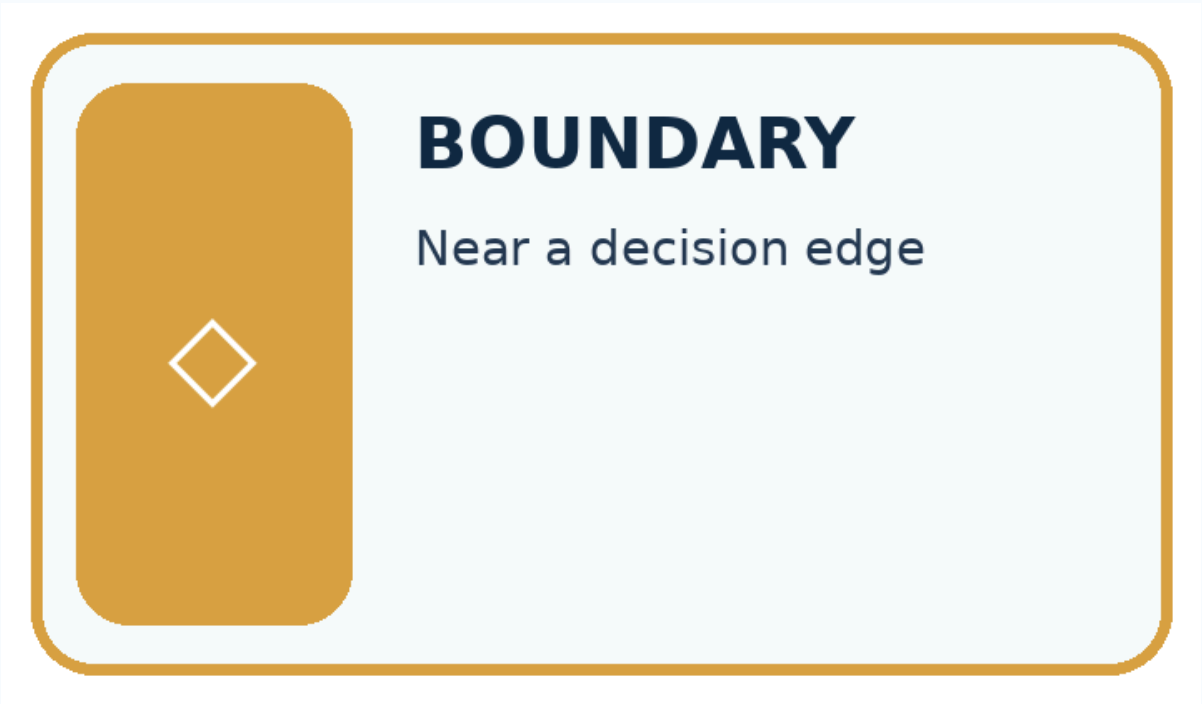
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

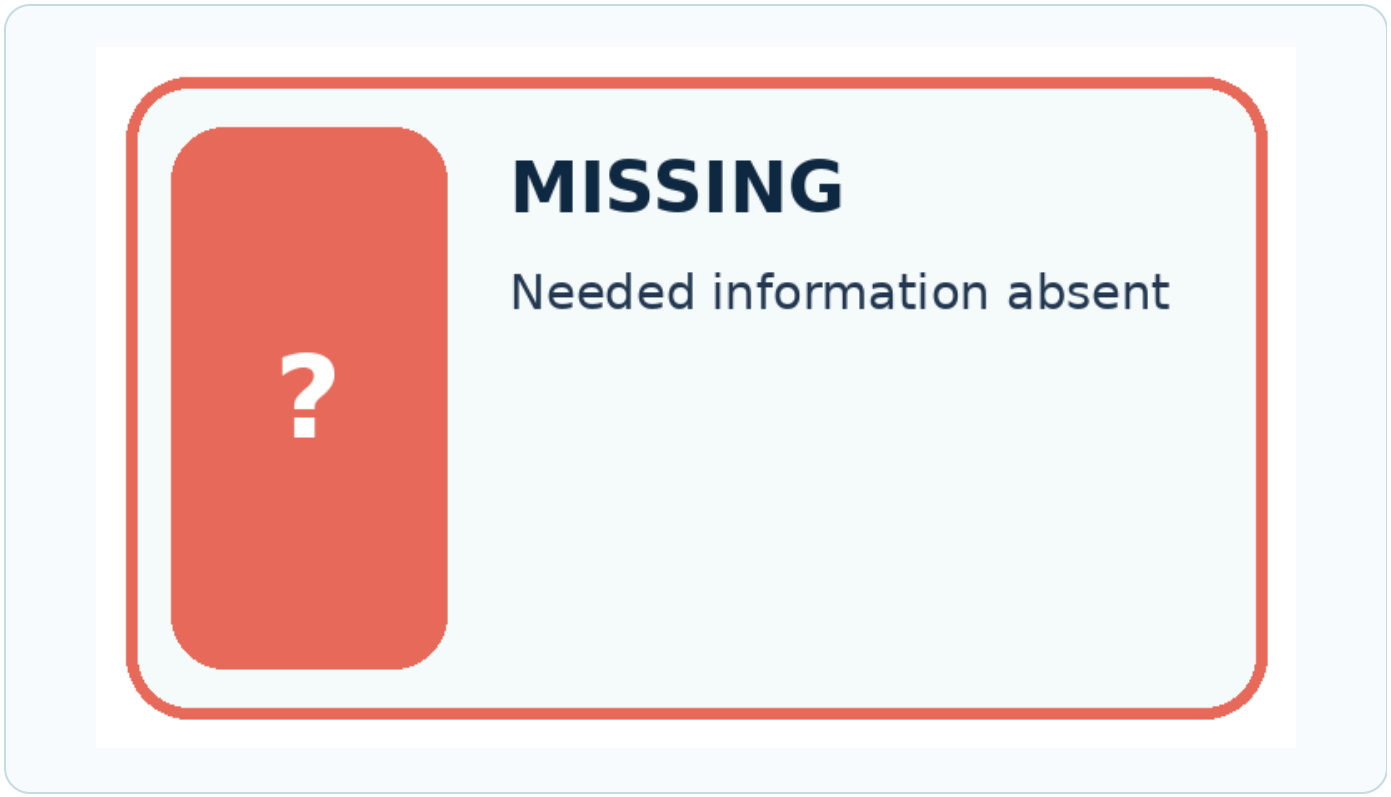
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Invite only the 15 library-club members instead.

Predicted result

Prediction: The invitation **process** no longer covers the whole grade through this frame.

Why the result follows

Reason: Club members may differ in reading interests; selection is now limited to one group.

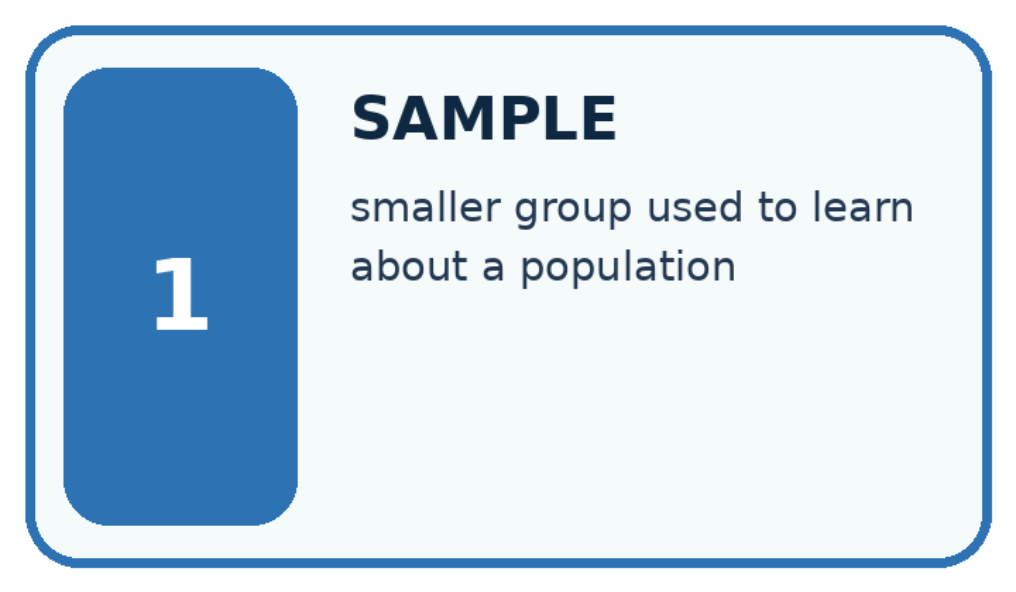
Words that unlock the idea

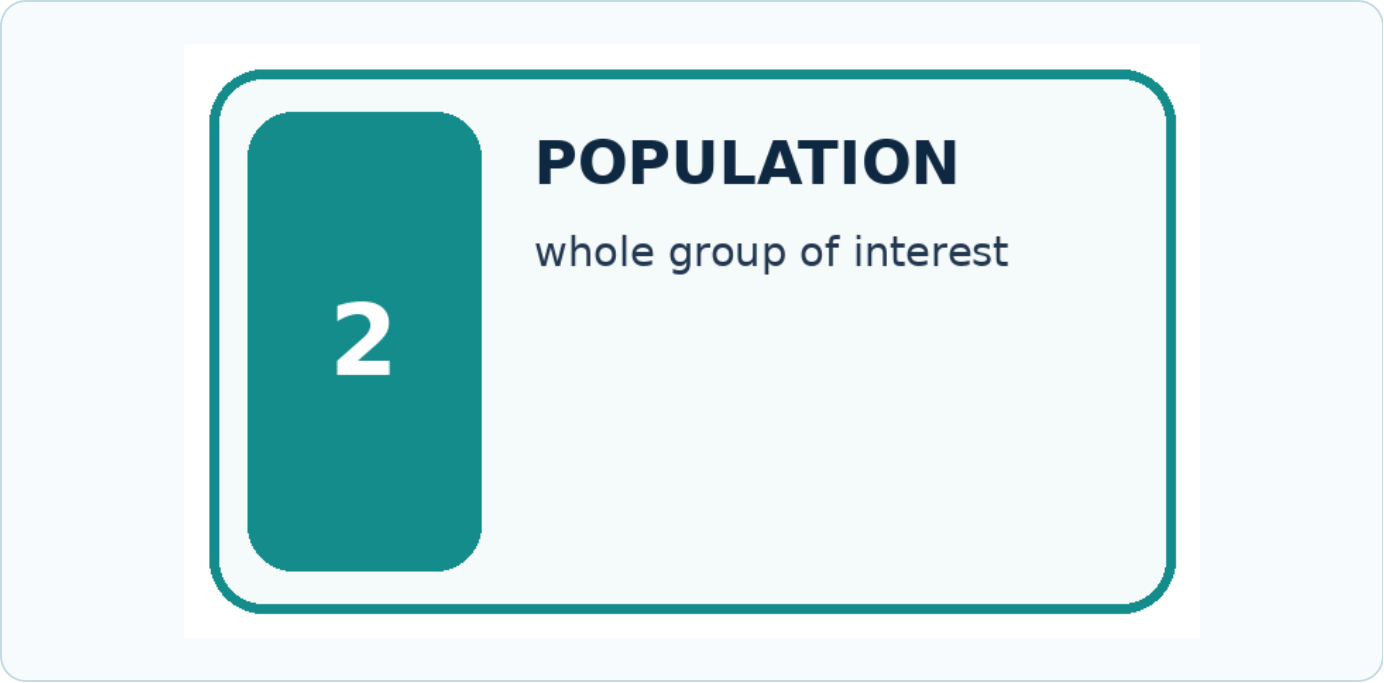
> Use the vocabulary of Sampling and Representation to explain a concrete decision, rather than only memorising definitions.

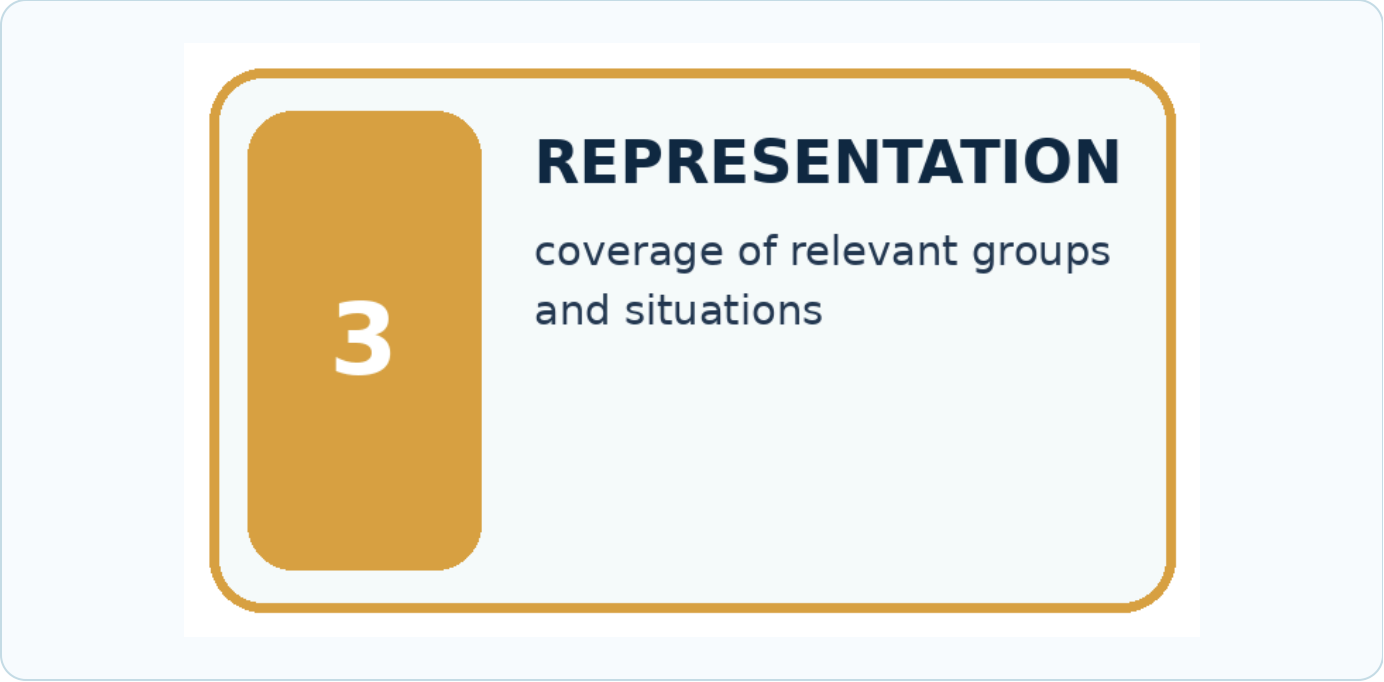
VOCABULARY LAB

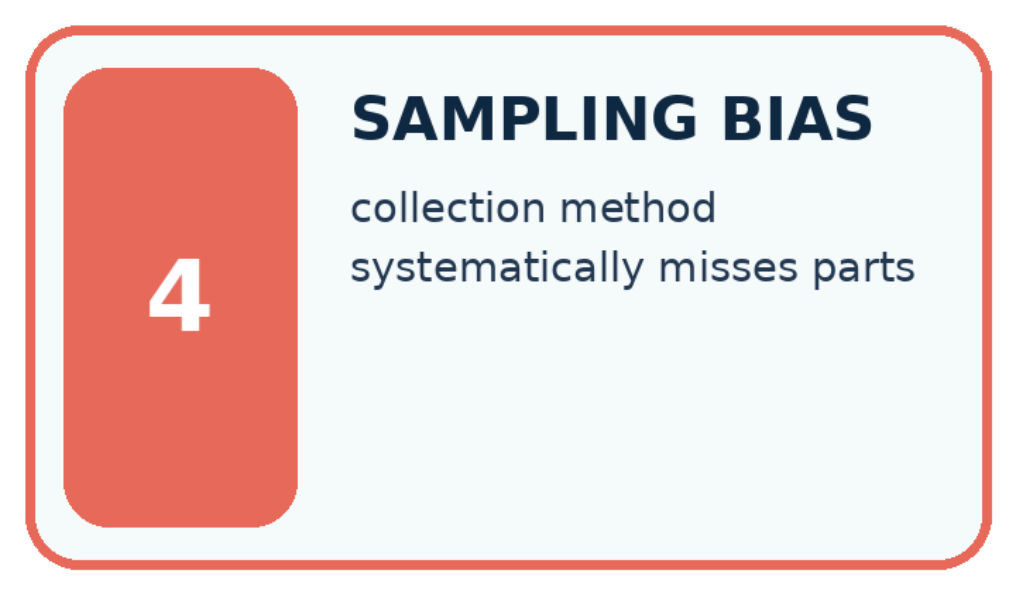
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
POPULATION
whole group of interest


3
REPRESENTATION
coverage of relevant groups and situations



Word	My own example	Not this / boundary
sample	A selected group of observations, or an outcome drawn from a distribution; the context identifies which meaning applies.	
population	The full group of cases or people a study aims to describe.	
representation	A way to encode information so a person or system can work with it.	
sampling bias	A systematic mismatch between selected examples and the population the study intends to describe.	

Words used in the closer explanation

Word	Meaning
Sampling frame	The list or source from which a sample is selected
Response	A recorded answer from a selected participant

Terms for Graphs and honest scales

Term	Meaning
Axis	A reference line with a scale on a graph
Baseline	The starting reference value on a scale
Bar chart	A graph using bar lengths to compare categories
Scale	The mapping from values to positions or lengths

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Sampling and Representation, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 19 • WORKED EVIDENCE

1. Goal: books for the whole school.
2. Sample: one advanced-reading club at lunchtime.
3. Missing: younger readers, other interests, absent students, different access needs.
4. Repair collection across grades, times, interests, and access needs; then recheck.

1. Goal: books for the whole school.

2. **Sample**: one advanced-reading club at lunchtime.

3. Missing: younger readers, other interests, absent students, different access needs.

4. Repair collection across grades, times, interests, and access needs; then recheck.

Recalculate, retrace, or restate the reasoning in your own **representation**:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

If the school adds 20 more club responses, which coverage problem remains?

Apply the connected idea: Graphs and honest scales

Draw bars for counts 4 and 6 using one square per vote. State the difference and ratio.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sampling and Representation

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

The facts we will use

Supplied facts

Population: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Supplied facts

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

AI ACADEMY / AI-2/19

Sampling and Representation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define target

The question concerns all Grade 7, not only library-club members.



2. Select invitations

Teacher uses the complete approved list and documented random selection.



3. Record responses

12 replies from 15 invited; three nonresponses remain visible.



4. Report scope

Invitation response rate 12/15=80%; preference counts describe respondents.

CHANGE ONE THING

Invite only the 15 library-club members instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define target	The question concerns all Grade 7, not only library-club members.
2. Select invitations	Teacher uses the complete approved list and documented random selection.
3. Record responses	12 replies from 15 invited; three nonresponses remain visible.
4. Report scope	Invitation response rate 12/15=80%; preference counts describe respondents.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

What this example does not establish

Keep the stated limit

Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

Keep the stated limit

Read the labelled picture: Can **80% response** be reported as **80%** of the whole grade responding?

Keep the stated limit

Change one thing: Invite only the 15 library-club members instead. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Sampling and Representation claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A large sample from one narrow group represents everyone. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Selection and **response** are separate. Even a fair invitation **sample** can be distorted if only interested students respond. Report numbers invited and responding and explain who may be absent.

> **Twenty students are selected fairly but only 8 answer. Does fair selection remove the need to report nonresponse?**

Apply the connected idea: Graphs and honest scales

Counts alone do not show why students chose an activity or whether nonrespondents would choose differently. Graph choices should match the question: categories suit bars; ordered time **observation**s can suit a line.

> **Art has the most votes among 18 responses. Does that establish Art is best for learning?**

STUDENT LESSON · 8 OF 18

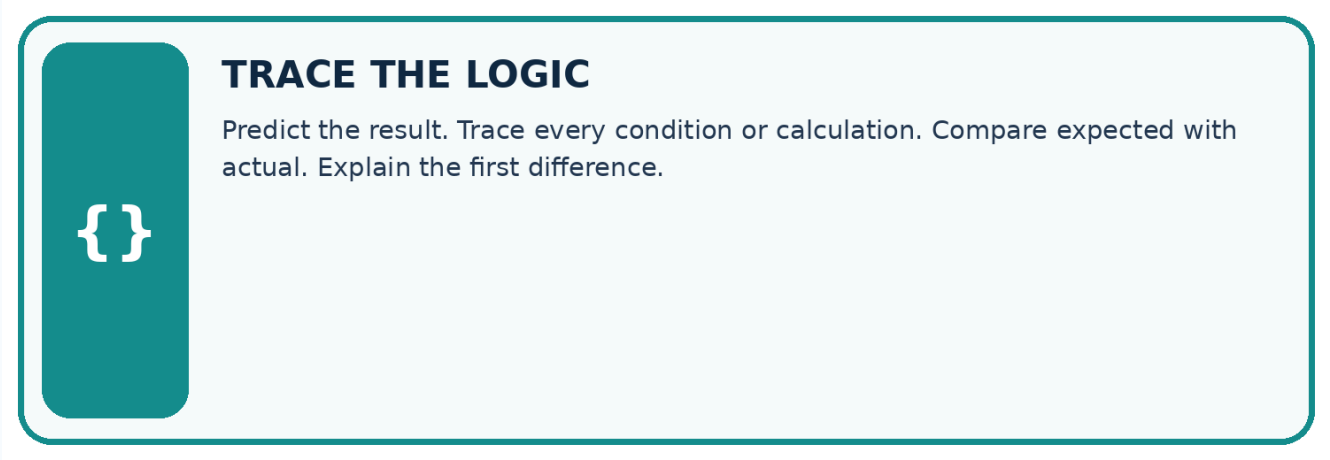
Practise together

> Practise Sampling and Representation with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Changing the report heading from club **response**s to all students does not fix sampling. Change the collection plan or narrow the claim.

Avoid collecting names when anonymous counts are sufficient; a teacher can manage selection without publishing identities.

Apply the connected idea: Graphs and honest scales

A graph with bars beginning at 7 turns counts 8 and 10 into visible lengths 1 and 3. Redraw from zero and label counts directly.

A line chart can use a restricted **axis** for detail if clearly labelled, but it must not disguise its **scale**.

Practise with less help: Read the sample result

Use workbook W2 for the connected example "Questions **population**s and **sample**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Questions **population**s and **sample**s. Proposed explanation: "A successful run proves the conclusion". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Graphs and honest scales

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

> Can **80% response** be reported as **80%** of the whole grade responding?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

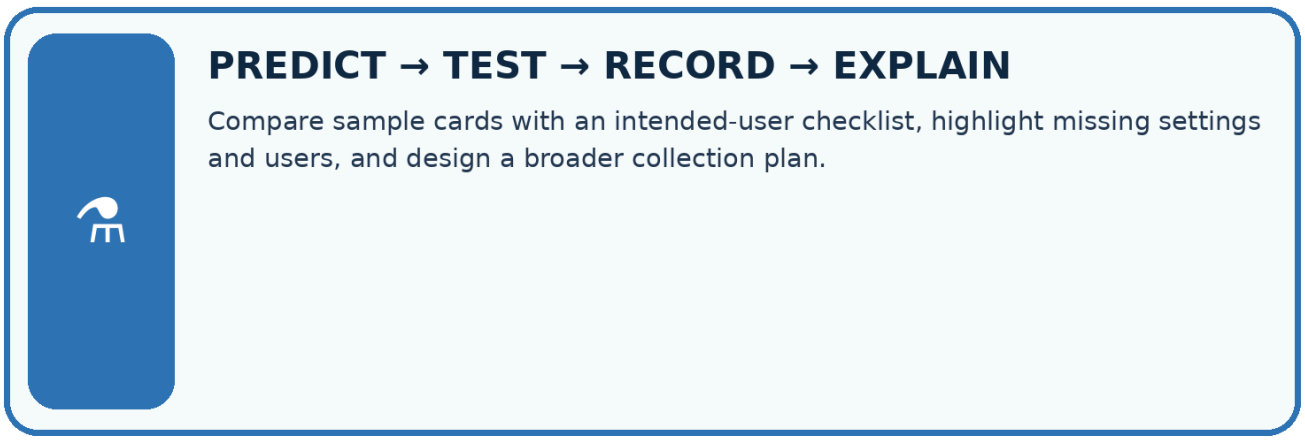
Run the investigation

> Investigate Sampling and Representation. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Compare **sample** cards with an intended-user checklist, highlight missing settings and users, and design a broader collection plan.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

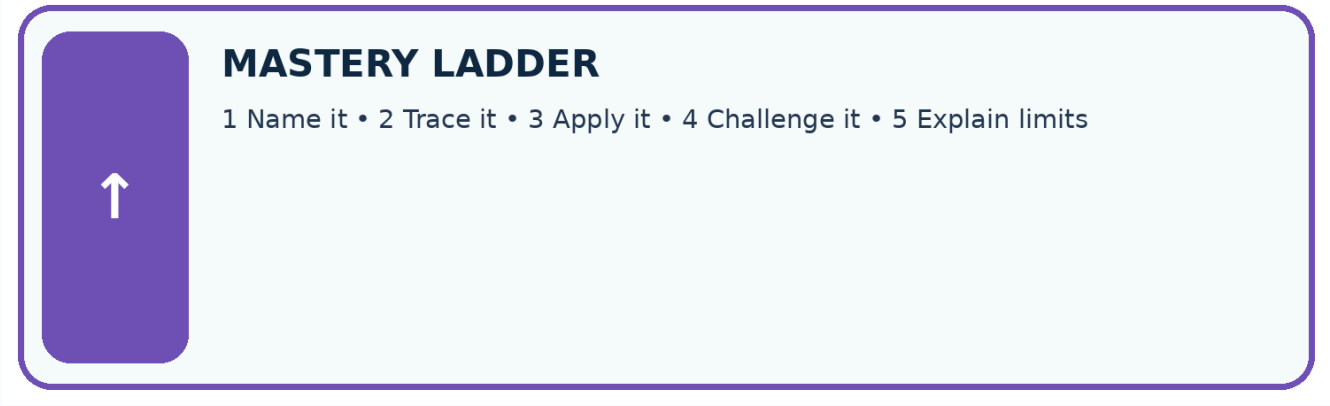
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Sampling and Representation. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

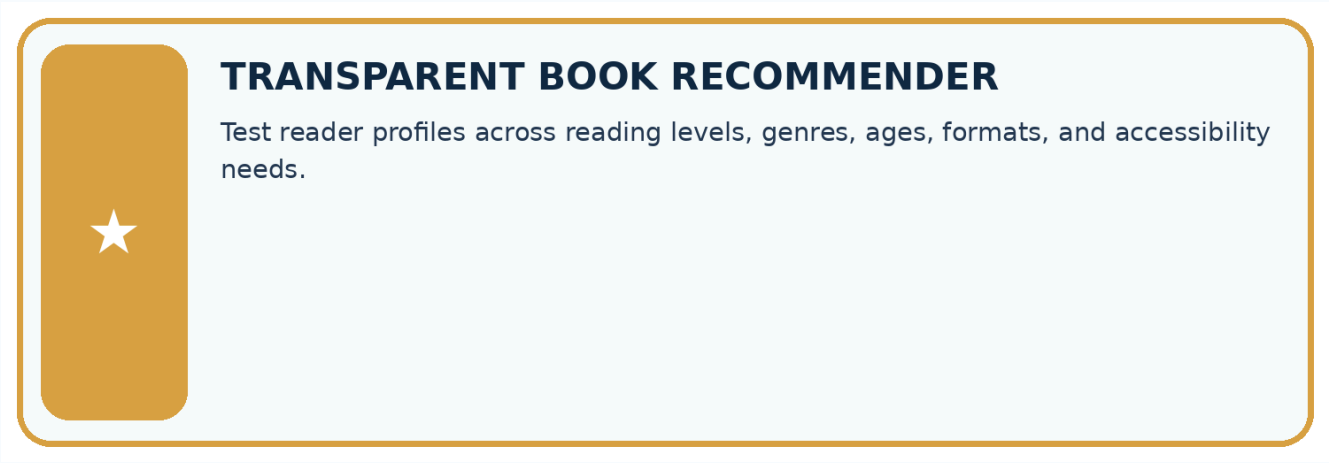
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

STUDENT LESSON · 12 OF 18

Show what you know

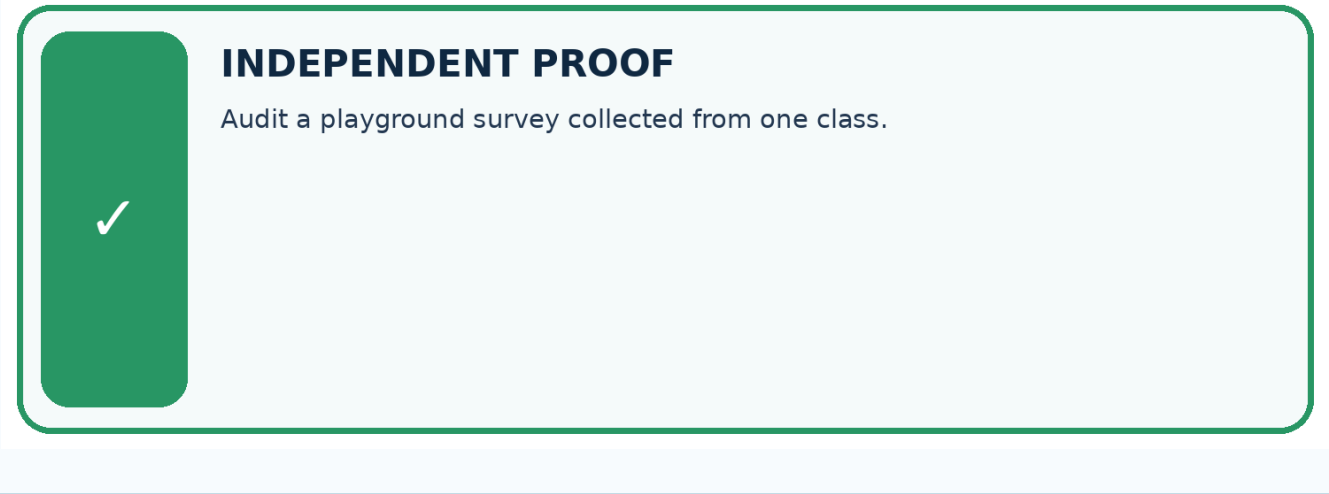
> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Audit a playground survey collected from one class.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Sampling and Representation case for supported practice. Repeated examples do not become fresh assessments.

Week 19 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Only adventure-club members answer a school reading survey.

The question

Explain the sampling problem.

Worked explanation

Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Who was left out of the survey?

Evidence or stage	Value or result
Sample	Adventure-club members
Intended population	School readers
Missing perspectives	Other interests and relevant grades

Only adventure-club members answer a school reading survey.

Explain the sampling problem.

Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Plan independently

Design an anonymous **sample** plan to ask all Grade 7 about three activity choices. State **population**, **sampling frame**, selection method, question, and how you will report non**response**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 19 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 **response**s may still fail to represent the school. Propose a better sampling plan using fictional data.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Sampling and Representation, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

By the end, I can	How I will show it
Define a question and target population	Explain the deciding step, show a trace or test, and state one limit.
Describe how a sample was selected	Explain the deciding step, show a trace or test, and state one limit.
Limit conclusions to supported groups	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Sampling and Representation without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Define a question and target **population**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

1. Define target

2. Report scope

3. Record **response**s

4. Select invitations

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report scope Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Invite only the 15 library-club members instead. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The invitation **process** no longer covers the whole grade through this frame.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The invitation **process** no longer covers the whole grade through this frame. Club members may differ in reading interests; selection is now limited to one group.

Repair a missing building block

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Over**fitting** Intuition (ai-2/16). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does zero **training** error justify choosing Complex in the original record?.

Return to this week and explain what you changed.

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

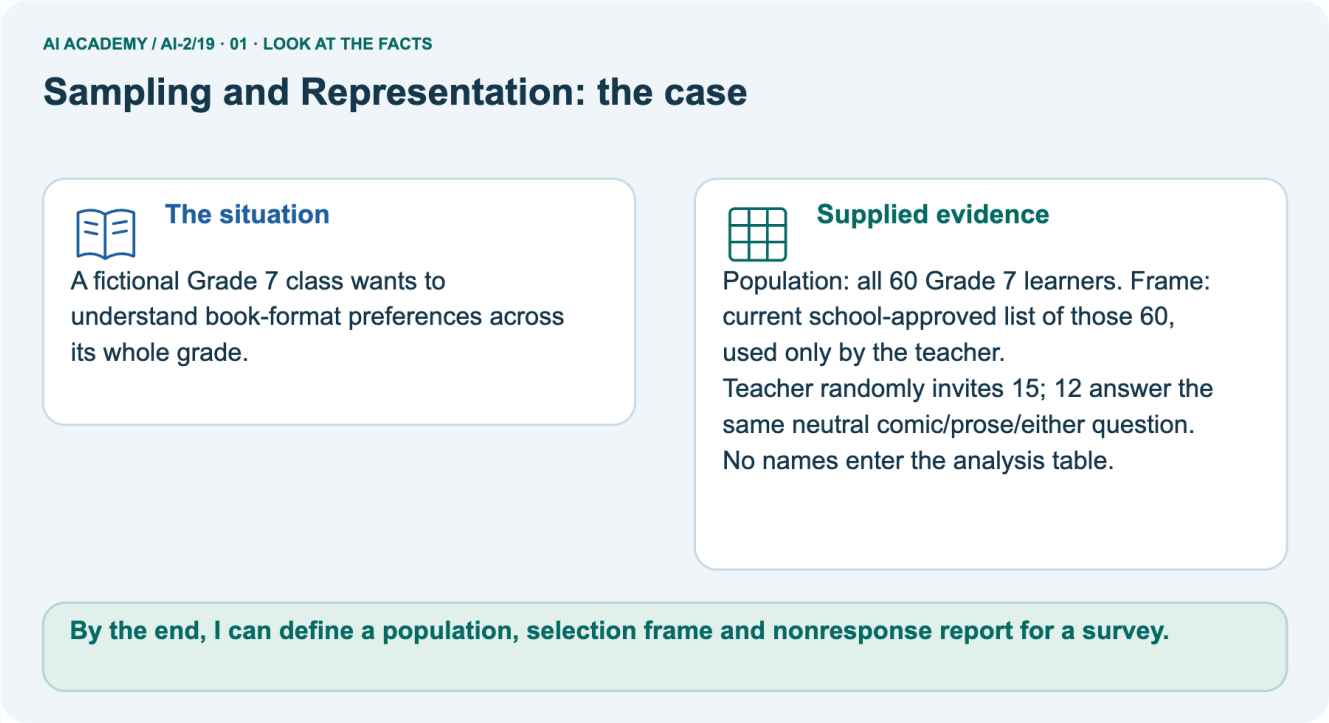
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Sampling and Representation graphic using its labels, deciding evidence and relevant limitation.

01. Sampling and Representation: the case



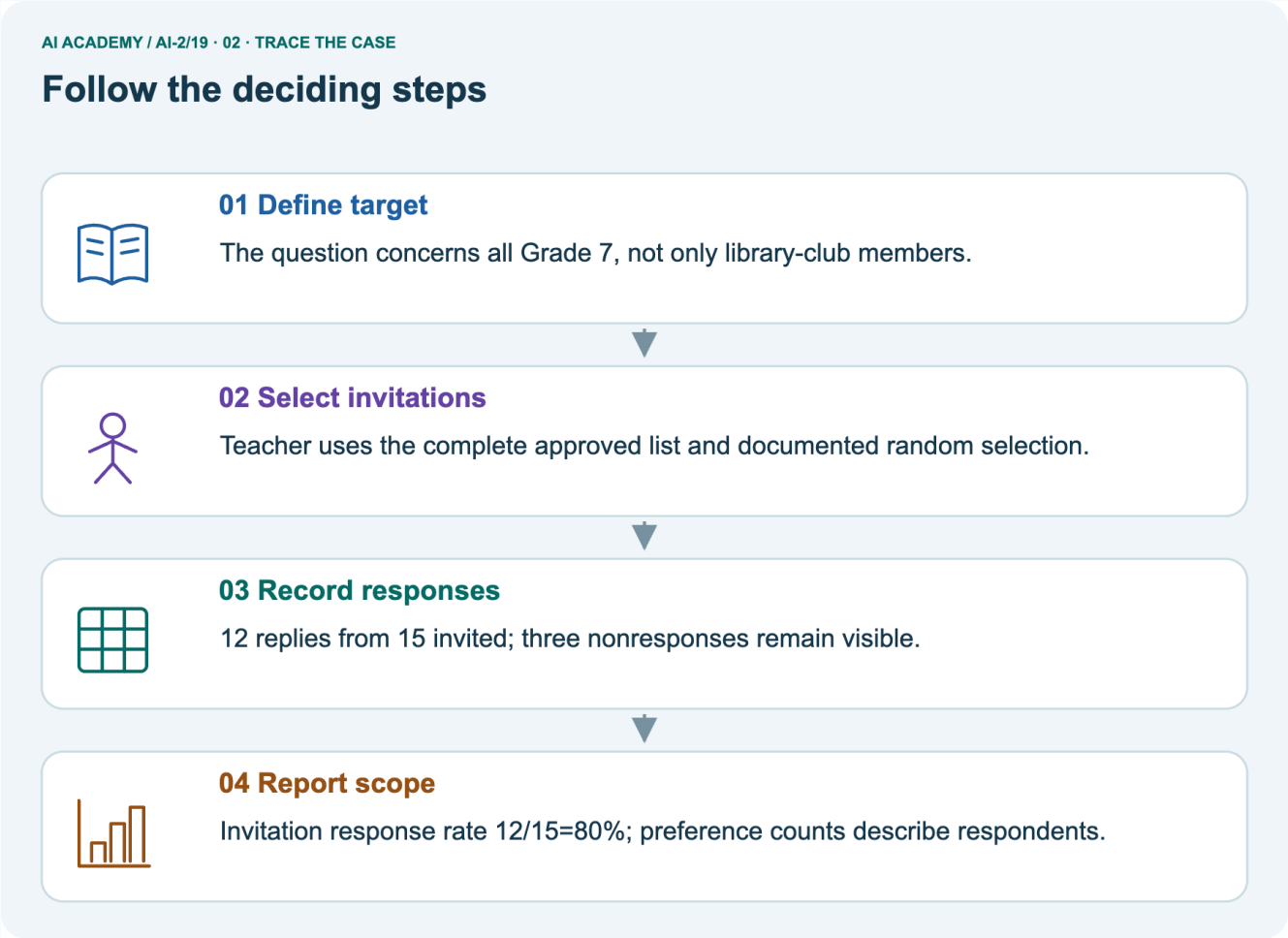
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Grade 7 class wants to understand book-format preferences across its whole grade. Supplied facts: **Population**: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

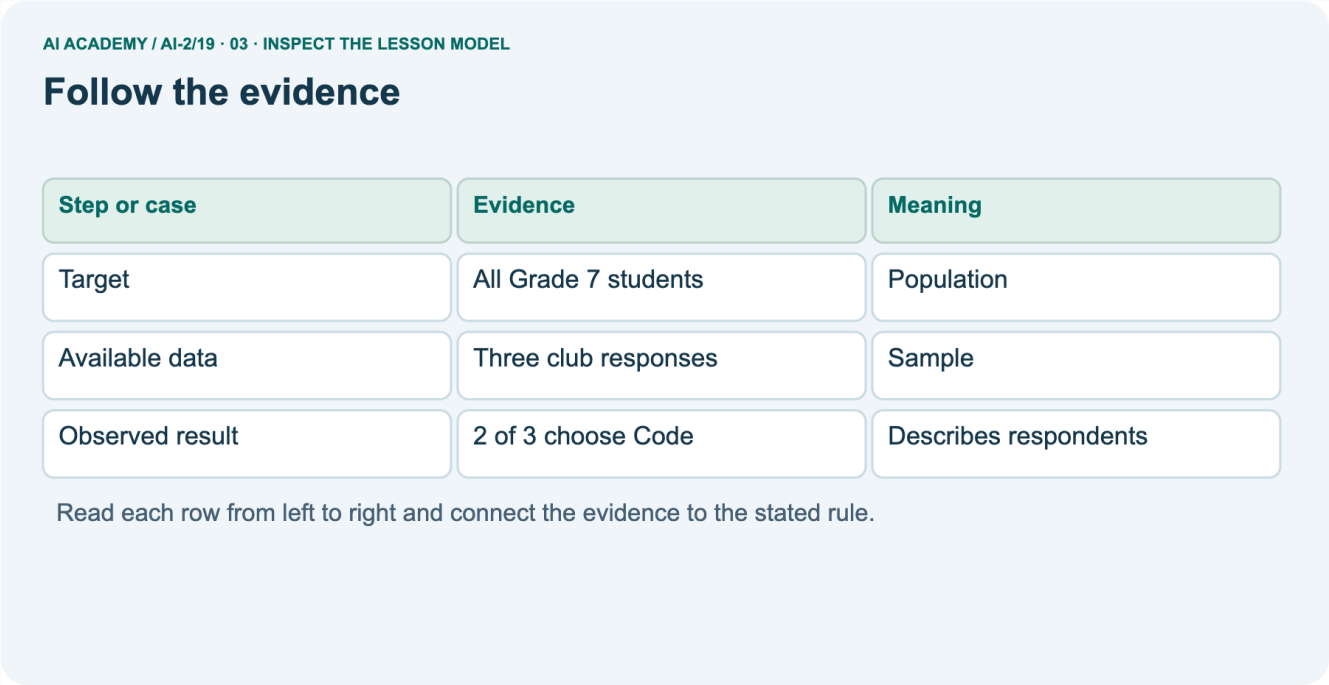


Follow the labelled picture with the matching explanation. Open the vector diagram

Define target: The question concerns all Grade 7, not only library-club members. → Select invitations: Teacher uses the complete approved list and documented random selection. → Record **response**s: 12 replies from 15 invited; three non**response**s remain visible. → Report scope: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

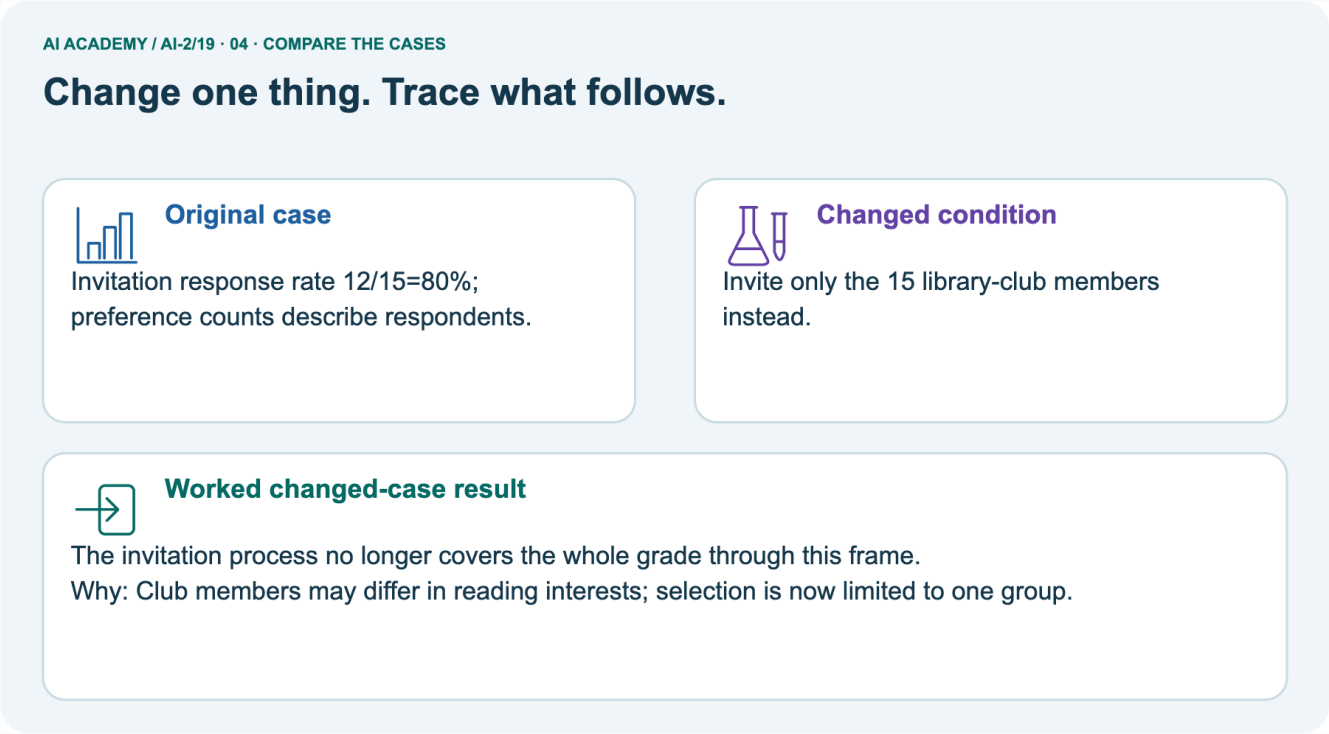


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Target | All Grade 7 students | **Population**. Available data | Three club **response**s | **Sample**. Observed result | 2 of 3 choose Code | Describes respondents

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



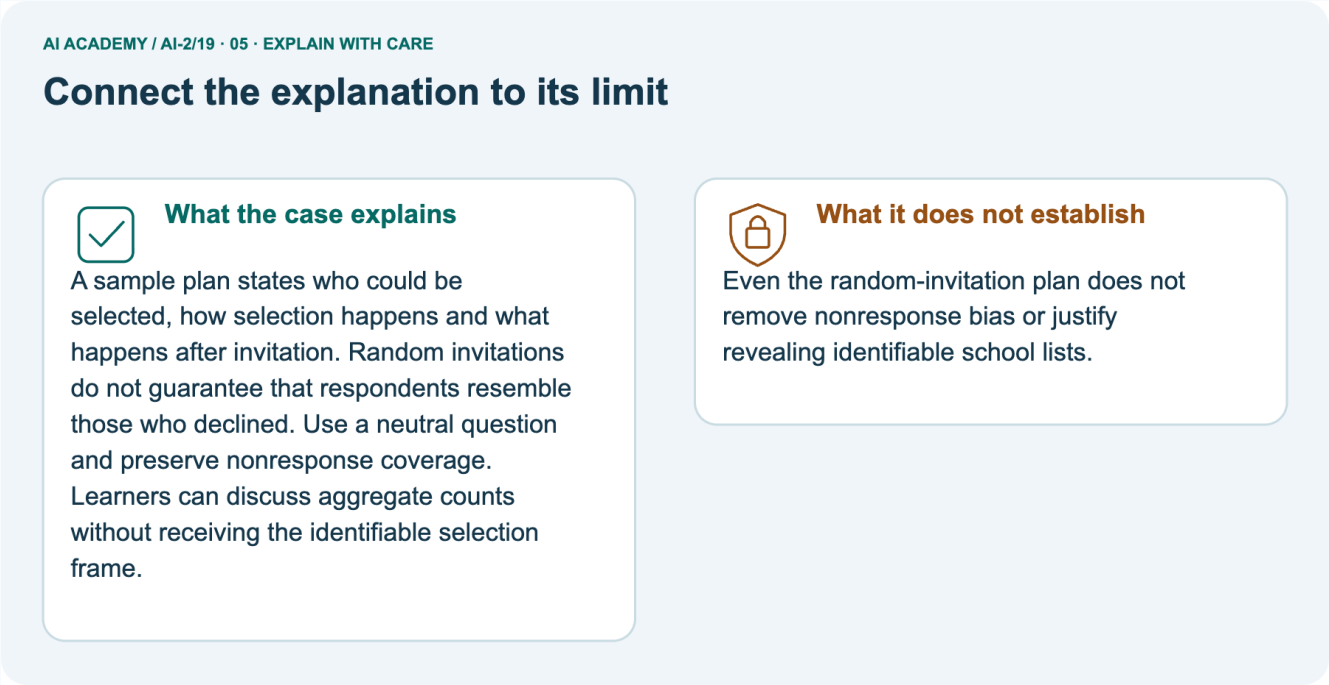
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.
Changed condition: Invite only the 15 library-club members instead.

Expected result: The invitation **process** no longer covers the whole grade through this frame.
Reason: Club members may differ in reading interests; selection is now limited to one group.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

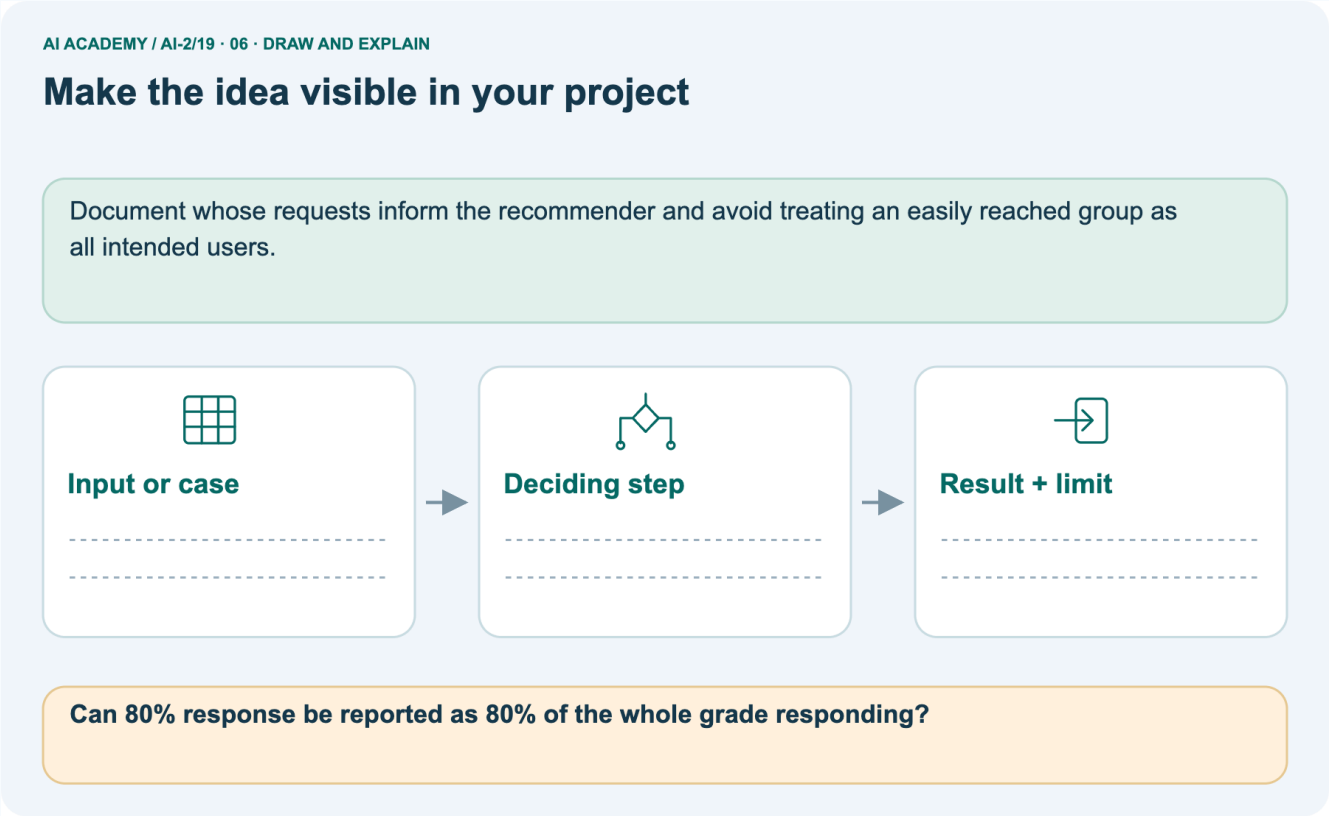
A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Limit: Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users. Draw your **input**, deciding step, result and limit.

Supported question: Can **80% response** be reported as **80%** of the whole grade responding?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.